



Process Management

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Operating System Concepts

■ Definitions

- The term process was used by the designer of MULTICS system in 1960.
- A program in execution.
- The animated spirit of a procedure.
- Entity to which processor is assigned.
- It is the dispatchable unit.

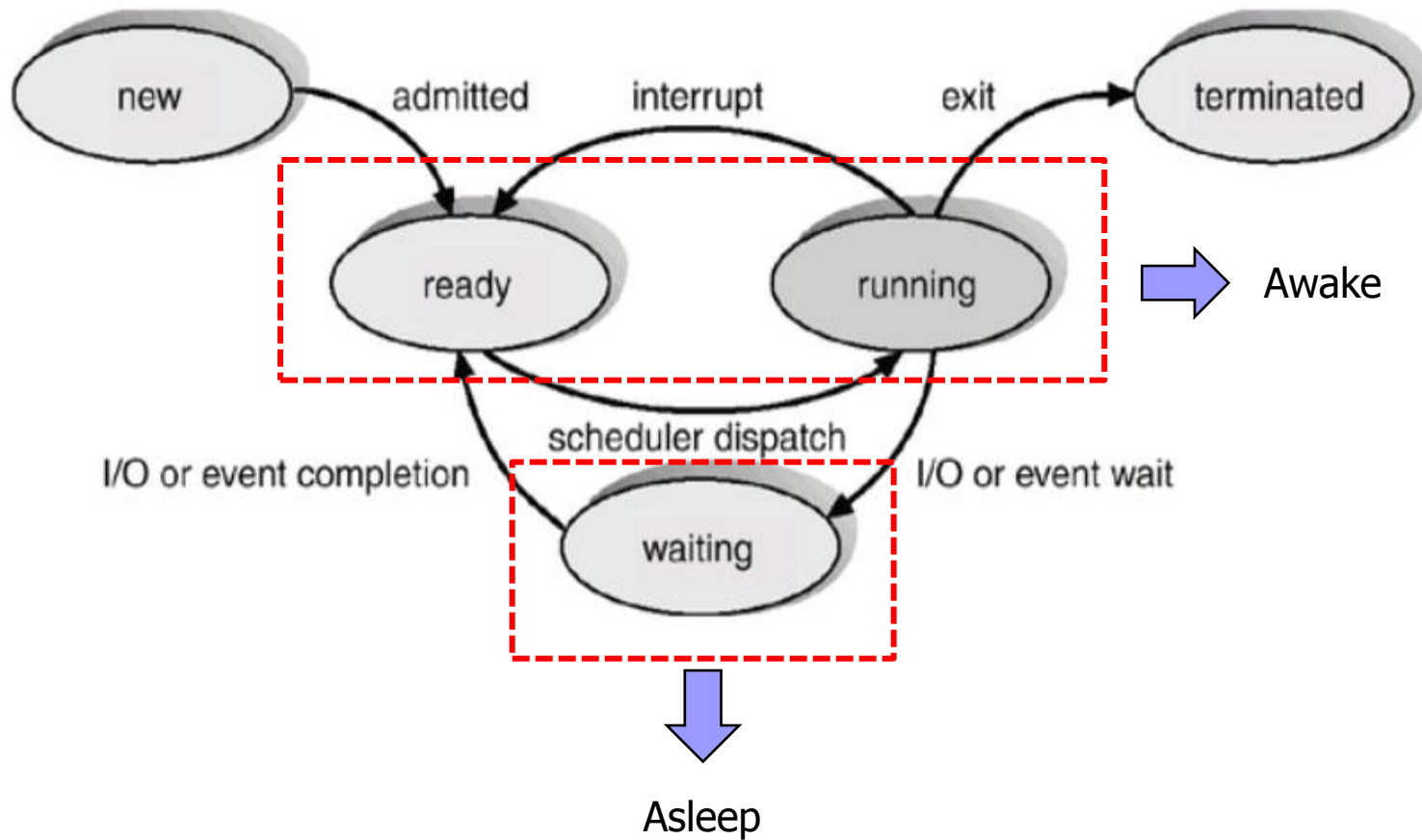
- Each process has its own address space, which typically consists of,
 - Text region
 - Code of the program that CPU executes
 - Data region
 - It stores variables and dynamically allocated memory that the process uses during execution.
 - Stack region
 - It stores instructions and local variables for active procedure calls.

Process states, life cycle of a process

■ Process states

- The OS must ensure that each process receives a sufficient amount of CPU time.
- During its life time a process moves through a series of discrete process states.
- Various events can cause a process to change state.
- There are three states particularly,
 - New
 - Ready
 - Running
 - Blocked
 - terminated

Process states



Process states

- **New** → The process is being created
- **Running** → Instructions are being executed
- **Waiting** → The process is waiting for some event to occur (I/O or reception of signal)
- **Ready** → The process is waiting to be assigned to a processor
- **Terminate** → The process has finished execution

■ Running

- A process is said to be running if it is executing on a processor.

■ Ready

- A process is said to be ready if it could execute on a processor if one were available.
- Process waiting state before getting the CPU.

■ Blocked

- A process is said to be in blocked state if it is waiting for some event to happen before it can proceed.
- E.g. I/O completion event.

Process State Transitions

- Changing states during execution of a process
- Following transitions are possible during the complete execution of a process,
 - Dispatch
 - Timerruout
 - Block
 - Wakeup

State transitions

■ Dispatch

- The assignment of the CPU to the first process on the ready list is called dispatching.
- It is performed by a system entity called dispatcher.
- We indicate the transition as follows,

Dispatch (process name): ready → running

■ Timer runout

- If a running process does not voluntarily leave the CPU before the time interval expires, the interrupting clock generates an interrupt, causing the operating system to regain control.
- The OS then makes the previously running process ready and makes the first process on the ready list running. Transition indicated as

Timerrunout (process name): running → ready
and

Dispatch (process name): ready → running

■ Block

- If a running process initiates an input/output operation before its quantum expires, the running process willingly leave the CPU and wait in the blocked state until the desired event get.
- Transition is,

Block(process name): running → blocked

■ Wakeup

- When an input/output operation completes for which this process was in pending state.
- This state transition is initiated by the user process itself is block, the other three transitions are initiated by the operating system.
- The transition take place,

Wakeup(process name): blocked → ready

PCB (Process Control Block)

- The OS typically performs several operations when it creates a process.
- First it uniquely identifies each process by assigning a PID.
- Secondly, OS creates a PCB/PD (process descriptor), which maintains information that the OS needs to manage the process.
- PCB basically is data structure in the OS kernel which hold certain information about a process.

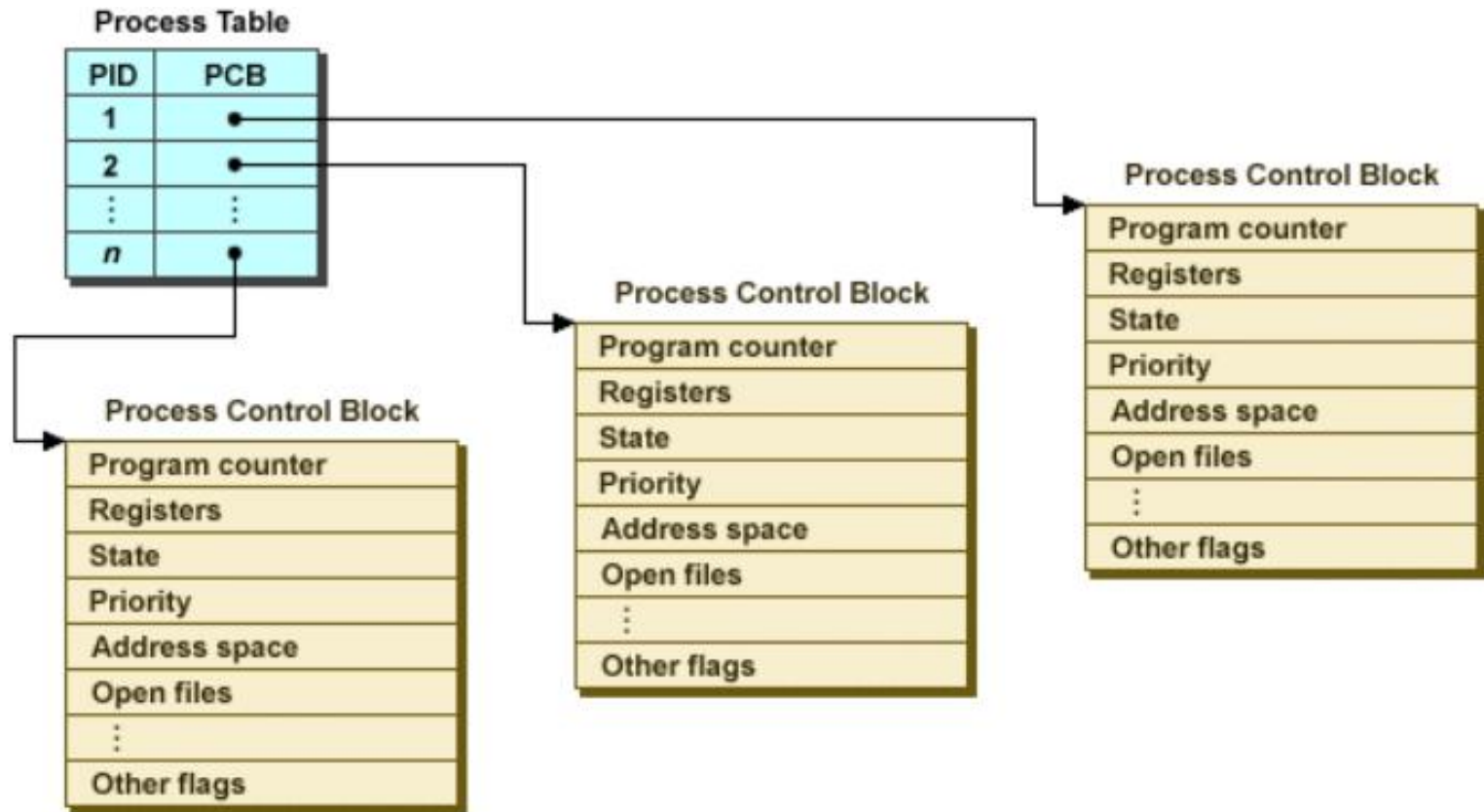
PCB (Process Control Block)

- The information that PCB has,
 - PID
 - Process state
 - Program counter
 - Scheduling priority
 - Credentials (data that determines the resources this process can access)
 - A pointer to the process's parent process
 - Pointers to the process's child process.
 - Pointers to locate the process's data and instruction in memory
 - Pointers to allocate resources

PCB (Process Control Block)

- The **PCB** also stores the register contents, called the execution context, of the processor on which the process was last running, when it transitioned out of the running state.
- i.e. General purpose registers data.
- When a process transitions from one state to another, the OS must update information in the process's PCB.

Process Table and PCB



Process Table

The Process Table is a data structure maintained by the operating system to facilitate context switching and scheduling, and other activities.

Process operations

- OS must be able to perform certain process operations, including:
 - Create a process
 - Destroy a process
 - Suspend a process
 - Resume a process
 - Change a process priority
 - Block a process
 - Wake up a process
 - Dispatch a process
 - Enable a process to communicate with other processes (IPC)

Process Operations

- A process may spawn a new process. If it does
 - The creating process is called parent process.
 - The created process is called child process.
 - Each child has exactly one parent but each parent may have more than one child.
 - Process creation in UNIX and Linux are done through `fork()` or `clone()` system calls.

System Call

In computing, a **system call** is how a program requests a service from an operating system's kernel.

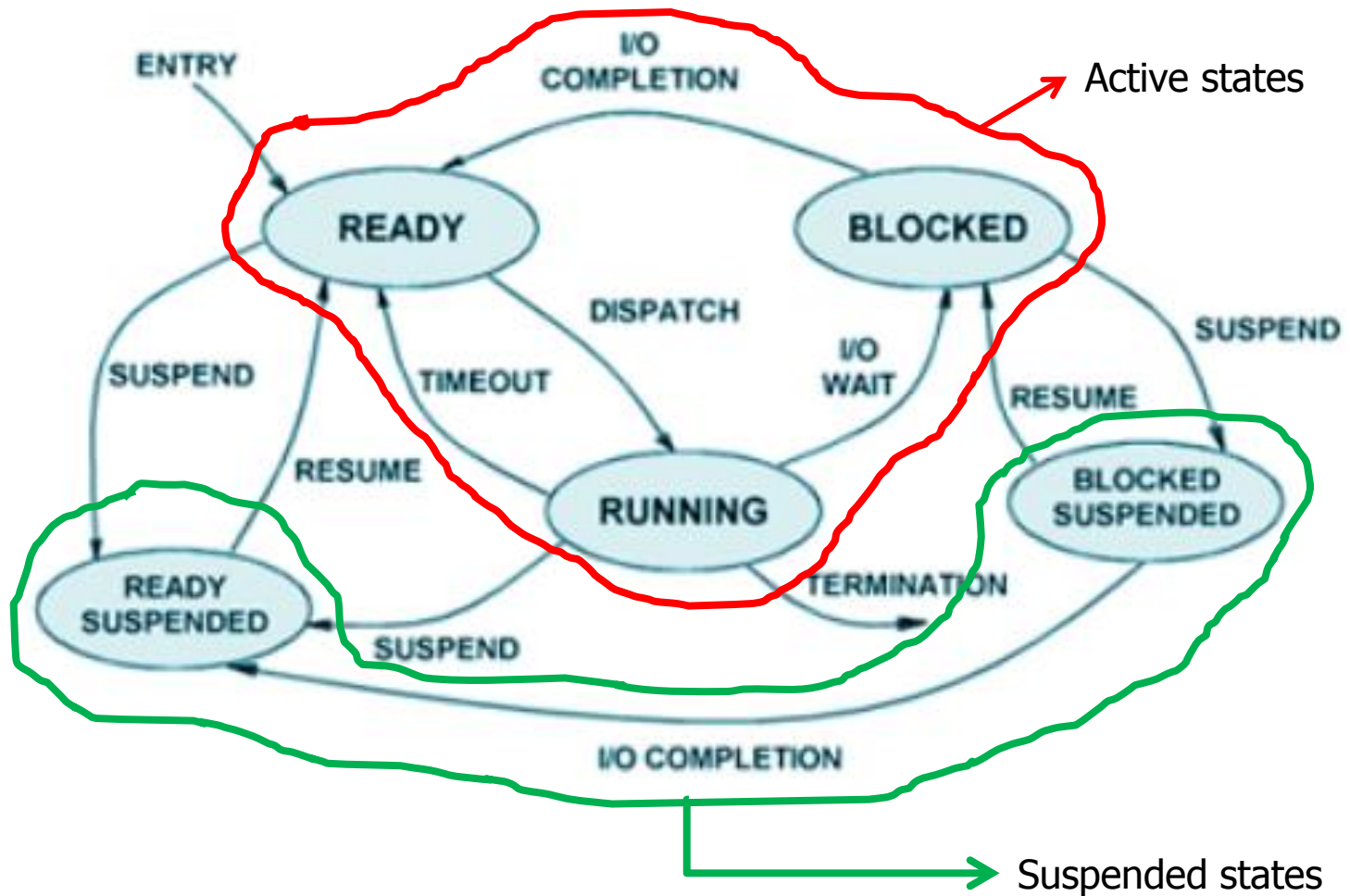


State model with suspend & Resume

- A process may be swapped out of working memory by the OS's memory manager in order to free up memory for another process.
 - When a process is suspended, it essentially becomes dormant until resumed by the system.
 - Because a process can be suspended while it is either ready or blocked, it may also exist in one of two further states,
 - Ready suspended and
 - Blocked suspended

- These operations are important for several reasons,
 - If a system is functioning poorly and may fail, then current processes may be suspended to be resumed after the problem is corrected.
 - A user suspicious about the partial results of a process may suspend it until the user can ascertain whether or not the process is functioning correctly.
 - In short term fluctuations in system load, some process may suspended and resume later when the load settles back to normal level.

State model with suspend & Resume



Five state model

- New transitions are,
 - Suspend(process name): ready → suspended Ready
 - Resume (process name): suspended Ready → Ready
 - Suspend(process name): blocked → suspendedblocked
 - Resume(process name): suspendedblocked → blocked.
 - Completion(process name): suspendedblocked → suspendedready

